Sahaj Bhadja

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Summary

Software engineer with 2+ years building **C++ simulations**, Python services, data pipelines, and tested cross-platform systems; experienced translating requirements into measurable solutions using version control, debugging, analytics, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Designed **normalized relational schemas and data services** for an AI platform supporting 500+ users, improving dashboard performance from 1.2 seconds to 350 milliseconds through SQL query, indexing, and access-pattern optimization.
- Built **ETL-style data pipelines** across FastAPI, SQL Server, PostgreSQL, and cloud storage, processing 5,000+ monthly jobs with reconciliation checks, error handling, and CI/CD-backed releases achieving 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Translated **instructor requirements** into React and Flask application features for uploads, dashboards, and grading workflows, supporting 150+ users and 10,000+ daily requests while maintaining stable production performance.
- Designed **concurrent batch processing** with multithreading and secure PDF, DOCX, and ZIP ingestion, processing 1,000+ files per batch through fault-tolerant execution, validation, testing, and REST APIs.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built a **GPT-based Decision Transformer** for offline RL, improving long-horizon trajectory planning 18% in simulation while developing reproducible data and evaluation pipelines processing 100K+ large-scale trajectory records.
- Built an **HD-map simulation encoder** combining road geometry and trajectory context through attention, improving closed-loop stability 15% while analyzing discrepancies across rigorous model evaluations and benchmark outputs.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Redesigned **OOP-based API and ETL components** using AWS Glue, S3, SQL, indexing, and caching, improving response times 30%, pipeline efficiency 68%, and maintainability across production workflows at scale.
- Automated **cross-platform builds and 20+ test suites** with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, reducing technical debt 20% while strengthening regression validation and release reliability.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Translated **business workflows into eight modular microservices** for authentication, resource allocation, customer records, billing, permissions, and notifications, using Azure SQL and WebSockets to synchronize 2M+ records across locations.
- Automated **Docker packaging, testing, and staged deployments** through Jenkins CI/CD, cutting release time 50% while improving consistency across development environments, code reviews, and reliable production rollout workflows.

Skills

Languages: **C++**, Python, Java, SQL, JavaScript, TypeScript, Bash, Shell
Software Engineering: OOP, Algorithms, Data Structures, Refactoring, Debugging, Requirements Analysis, Code Review, Git
Data & Analytics: PostgreSQL, MySQL, SQL Server, Azure SQL, Databricks, Spark, ETL, Data Mining, Statistical Analysis
Simulation: CPU Scheduling, Performance Optimization, Gantt Visualization, Data Processing, Feature Engineering, ML Models
Testing & Delivery: Linux, Docker, Jenkins, CI/CD, PyTest, Mockito, Unit/Integration Testing, Agile/Scrum

Projects

CPU Scheduling Simulator Live | *C++, React, Algorithms, Visualization*

- Implemented **six CPU scheduling algorithms** in C++ with React controls and Gantt visualization, validating 1K+ process combinations while reducing average turnaround time 25% and improving debugging efficiency 20% across simulated workloads.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Extended an **AI-powered SQL development workspace** supporting 10+ databases, improving query authoring and schema exploration across a Java and TypeScript codebase with Docker-backed development workflows on three operating systems.